

Data Flow Diagram

Date: 30 September 2026 **Team ID:** 6ab3e57da238dc999eb3d44d **Project Name:** EduGenie – AI-Powered Educational Assistant

Maximum Marks: 2 Marks

Current logical data flow

Entity / Process	Input	Processing	Output
Learner (External Entity)	Question, topic or passage	Submits selected learning task	Requested learning result
Web Interface (Process)	User text + selected tool	Sends JSON request to FastAPI	Displays returned result
FastAPI Router (Process)	Validated request	Routes to the appropriate module	Module-specific response
Feature Module (Process)	Topic/text	Builds learner-oriented prompt	Prompt to shared AI service
AIService (Process)	Prompt	Calls Gemini when configured; otherwise uses demo fallback	Generated text / quiz JSON
Gemini API (External Service)	Prompt	Generative inference	AI response
Quiz JSON validation (Data Processing)	Raw quiz response	Cleans/parses JSON and validates structure	Three-question quiz or error

Data stores: the current release has no persistent application database. Configuration such as GEMINI_API_KEY is kept in environment variables rather than learner data storage.